

How Open Source Technologies Can Add Security to IoT

The Internet of Things (IoT) enables billions of Web connected devices to exchange meaningful information without human intervention. It has the power to revolutionise the retail business, hospitality, healthcare and other sectors. To ensure IoT based devices are connected to each other properly, open source software now plays an increasingly important role.



Let us look at the example of an intelligent device like a camera that monitors the road for congestion, traffic accidents and weather conditions. In the world of IoT, the camera relays that information to a gateway, combines it with data from other traffic cameras and forges a citywide traffic monitoring system, that is in turn connected to other traffic systems. If a traffic camera detects a car accident on a main road, for instance, that information is sent to a central city traffic system to enable traffic to be rerouted in order to avoid the accident spot.

Open source software

Open source software, as we all know, can be used freely. Programmers write the code. Other people can use and change the code to a certain extent, depending on the licence. The process of changing the original code is known as upstreaming, and those who make the change become contributors. Resources and community websites are the main examples where such open source code is available free. Though misinformation can exist in such communities, the openness allows errors to be detected and corrected in real-time.

Open source software is closely related to IoT. Contemporary storage and processing devices cannot manage

Today, more devices are connected to the Internet than people. Cisco predicts that 50 billion devices will be connected to the Internet by 2020 and over 60 billion by the end of 2025, nearly five times the expected population at that time. The IoT helps these devices to connect to each other automatically.

IoT is transforming how individuals and organisations connect with customers, suppliers, partners and other individuals. It is all about connecting sensors, actuators and devices to a network and enabling the collection, exchange and analysis of the information generated.

Hardware innovations like the Raspberry Pi are making it easier, faster and cheaper to develop new devices. Networking standards for low power networks, like LoRaWAN or NB-IOT, create new opportunities for connecting very small devices to a network. New standards are being deployed specifically for IoT use cases. These include MQTT for messaging, OMA Lightweight M2M for device management, or W3C Web of Things and one M2M for service inter-portability. Finally, there has been significant improvement in data storage, data analysis and event processing, making it possible to support the amount of data generated in large scale IoT based systems.